

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI _ 590018



A Project Phase-1 Report on
“IoT and AI Powered Smart Agriculture Monitoring and Automation
System”

Submitted in partial fulfillment of the award of the
Degree of Bachelor of Engineering
in

ELECTRONICS & COMMUNICATION ENGINEERING

By

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Under the Guidance of
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JAIN COLLEGE OF ENGINEERING, BELAGAVI
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Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the project phase-1 entitled “**IoT and AI Powered Smart Agriculture Monitoring and Automation System**” presented by **Sushmita Bambalwadi (2JI23EC157), Swapnil Havale (2JI23EC158), Swati Shivakale (2JI23EC160), Vishwas Valmiki (2JI23EC182)** is bonafide student of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated for project phase-1 have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of project phase 1 prescribed for the Bachelor of Engineering degree.

Prof. Deepali. Patil

Project Guide

Dept. of E&CE, JCE Belagavi

Prof. V.R.Bagali

HOD, Dept. of E&CE

JCE, Belagavi

Dr. J. Shivakumar

Principal & Director

JCE, Belagavi

DECLARATION

I hereby declare that the project phase 1 work embodied in this report entitled "**IoT and AI Powered Smart Agriculture Monitoring and Automation System**" has been carried out by us at Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi, under the supervision of **Prof. Deepali Patil**. The report has not been submitted in part or full for the award of any degree of this or any other university.

To the best of our knowledge, the above statement made by the students, **Sushmita Bambalwadi(2JI23EC157), Swapnil Havale(2JI23EC158), Swati Shivakale(2J23EC160), Vishwas Valmiki(2JI23EC182)** can be accepted for partial fulfilment of the requirements for the degree of **Bachelor of Engineering**.

Name	USN	Signature
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Place : Belagavi

Date :



Jain College of Engineering, Belagavi

Dept. of Electronics and Communication Engineering

Vision of Institute

"To be a university as a resource of solution to diverse challenges of society by nurturing innovation, research & entrepreneurship through value based education."

Mission of Institute

- To provide work culture that facilitates effective teaching-learning process and lifelong learning skills.
- To promote innovation, collaboration and leadership through best practices.
- To foster industry-institute interaction resulting in entrepreneurship skills and employment opportunities.

Vision of Department

“To achieve excellence in education and research for developing globally competent, ethically sound Electronics & Communication Engineers”.

Mission of Department

1. To provide conducive environment for learning through structured student centric, teaching learning process.
2. To nurture needs of society by infusing scientific temper in students and to grow as a centre of excellence with efficient industry-institute interaction.
3. To inculcate self learning skills, entrepreneurial ability and professional ethics.

Program Educational Objectives (PEO's)

1. Graduates will be able to contemplate real-time social problems and deliver efficient solutions.
2. Graduates will be able to lead and succeed in professional careers.
3. Graduates will contribute through research and entrepreneurship.

Program Specific Outcomes (PSO's)

Graduates in the UG program in Electronics and communication engineering will be able to

1. Design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs..
2. Apply knowledge in various domain of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.



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PROGRAM OUTCOME'S (PO'S)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.



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11. **Project management and finance:** Demonstrate knowledge and understanding of the **engineering and management principles and apply these to one's own work, as a member and leader** in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Objectives :

1. Develop advanced research skills and methodologies to conduct in-depth investigations and analysis within specific academic domains or topics.
2. Refine oral and written communication abilities to articulate complex ideas, theories and findings persuasively and coherently to diverse audiences.
3. Foster critical thinking skills and interdisciplinary perspectives by evaluating scholarly literature, theories and methodologies and exploring connections across academic disciplines within the framework.

CO-PO/PSO Mapping:

L1: Remembering L2: Understanding L3: Applying L4: Analyzing L5: Evaluating L6: Creating

Course Outcomes	Description	Bloom's Cognitive level
BEC685.1	To develop skills in doing literature survey and preparing ppts for presentation.	L3
BEC685.2	To enable to acquire knowledge of topic and present it before panel of examiners and other students.	L3
BEC685.3	To develop skills of report preparation.	L3

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Strength of CO Mapping to PO/PSOs with Justification:

1: Slight (Low)

2: Moderate (Medium)

3: Substantial (High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3			3			2	2	3		2	3	3
CO2	3	3			3			2	2	3		2	3	3
CO3					3			2	2	3				

CO-PO-PSO	Justification
CO1 ⑦ PO1 (3)	Strongly mapped as students can understand how to do literature survey and prepare presentation.
CO1 ⑦ PO2 (3)	Strongly mapped as students can analyze, formulate, review research literature.
CO1 ⑦ PO5 (3)	Strongly mapped as students can create presentation and report using modern tools.
CO1 ⑦ PO8 (2)	Moderately mapped as students can apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
CO1 ⑦ PO9(2)	Moderately mapped as students can work in a team to prepare presentation.
CO1 ⑦ PO10(3)	Strongly mapped as students can write effective reports and design documentation , make effective presentation.
CO1 ⑦ PO12(2)	Moderately mapped as students can go for lifelong learning and higher studies in the field of chosen topic.
CO1 ⑦ PSO1(3)	Strongly mapped as students can use skills developed in engineering.
CO1 ⑦ PSO2(3)	Strongly can make use of knowledge gained in analog and digital electronics to present their seminar topic.
CO2 ⑦ PO1 (3)	Strongly mapped as students can apply the knowledge of engineering and science to present their seminar topics.
CO2 ⑦ PO2 (3)	Strongly mapped as students can analyze, formulate, review research literature.
CO2 ⑦ PO5 (3)	Strongly mapped as students can create presentation and report using modern tools.
CO2 ⑦ PO8 (2)	Moderately mapped as students can apply ethical principles and commit to professional ethics and responsibilities during their presentation.
CO2 ⑦ PO9(2)	Moderately mapped as students can work in a team to prepare presentation.



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CO2 7 PO10(3)	Strongly mapped as students can write effective reports and design documentation, make effective presentation.
CO2 7 PO12(2)	Moderately mapped as students can go for lifelong learning and higher studies in the field of chosen topic.
CO2 7 PSO1(3)	Strongly mapped as students can use skills developed in engineering.
CO2 7 PSO2(3)	Strongly can make use of knowledge gained in analog and digital electronics to present their seminar topic.
CO3 7 PO5 (3)	Strongly mapped as students can understand how to do literature survey and report preparation.
CO3 7 PO8 (2)	Moderately mapped as students can apply ethical principles and commit to professional ethics and responsibilities during their presentation.
CO3 7 PO9(2)	Moderately mapped as students can work in a team to prepare seminar report.
CO3 7 PO10(3)	Strongly mapped as students can write effective reports and design documentation, make effective presentation.

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ABSTRACT

Agriculture is one of the most important sectors that supports the economy and food supply of many countries. However, traditional farming methods mainly depend on manual observation and experience. Due to unpredictable climate changes, improper irrigation, soil degradation, and plant diseases, farmers often face crop loss and low productivity. To overcome these problems, modern technologies such as the Internet of Things (IoT) and Artificial Intelligence (AI) are introduced in agriculture. This project presents a Smart Agriculture System using IoT with the help of AI techniques. The system uses IoT sensors to continuously monitor soil moisture, temperature, and humidity in real time. The collected data is transmitted to a cloud server for storage and processing. AI algorithms analyze soil conditions and seasonal factors to recommend suitable crops. An AI-based image processing model detects plant diseases from crop images and suggests proper fertilizers and pesticides. A mobile application provides farmers with real-time data, crop recommendations, disease alerts, and government scheme information. The proposed system improves productivity, reduces manual effort, and supports sustainable farming.

CHAPTER 1

INTRODUCTION

Agriculture plays a vital role in ensuring food security and supporting the economy. However, traditional farming methods are increasingly facing challenges due to climate change, unpredictable environmental conditions, and growing population demands. Farmers often struggle with issues such as crop diseases, improper irrigation, and lack of timely information, which ultimately affect productivity and crop quality. To address these challenges, modern technologies like Internet of Things (IoT), Artificial Intelligence (AI), and Image Processing are being integrated into agriculture. This project focuses on developing a smart farming system that monitors crops in real time, detects diseases at an early stage, and provides accurate recommendations to improve overall crop health and yield.

1.1 MOTIVATION

The motivation behind this project arises from the limitations of existing farming practices. Farmers mainly depend on manual observation to identify crop diseases, which is not always accurate and may lead to late detection. Early-stage diseases are often difficult to recognize with the naked eye, resulting in severe crop damage if not treated on time. Additionally, improper use of water, fertilizers, and pesticides can further reduce productivity and increase costs. Most existing smart agriculture systems rely only on sensor data and lack visual analysis of crops. Therefore, there is a strong need for a complete and automated system that combines IoT, AI, image processing, and mobile technology to assist farmers in making better decisions.

1.2 PROBLEM STATEMENT

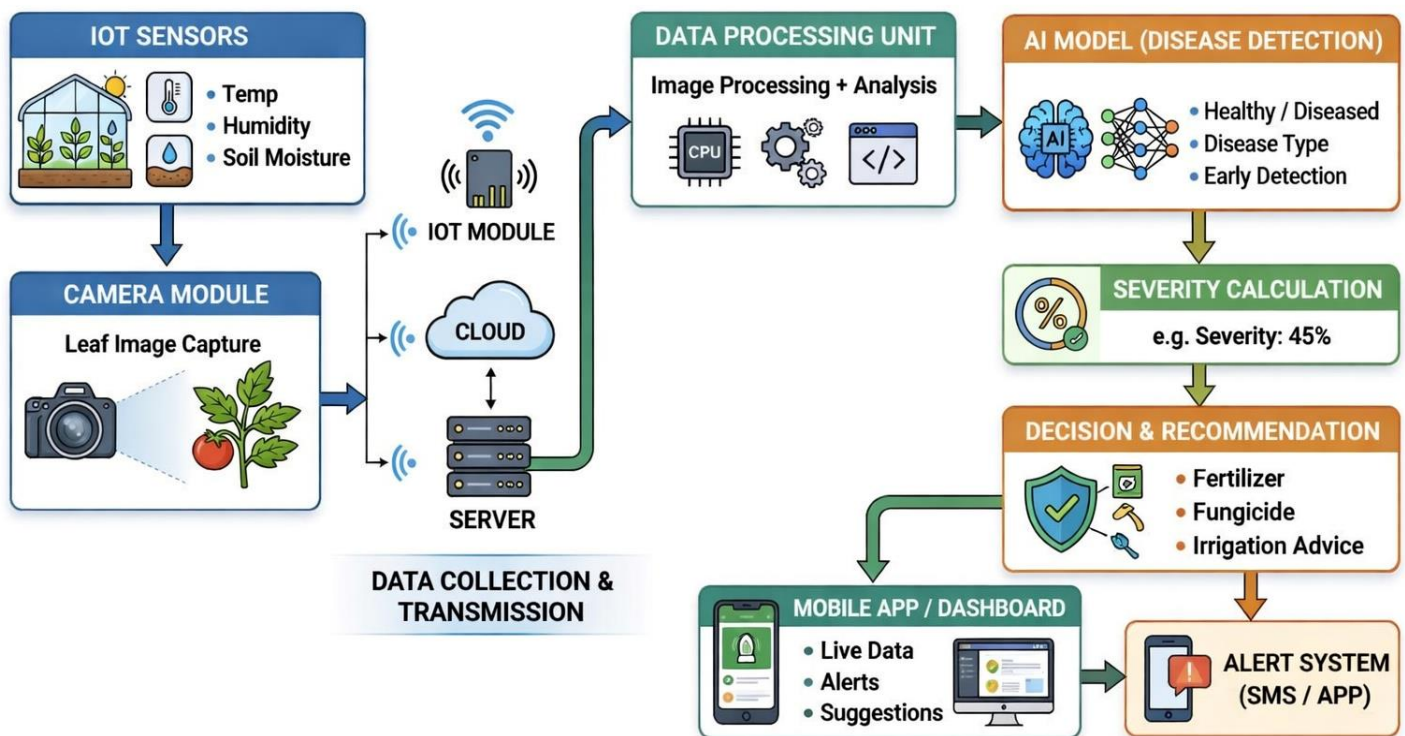
The main objective of this project is to design and develop a smart farming system that integrates IoT sensors, image processing, and AI techniques. The system aims to monitor environmental conditions such as temperature, humidity, and soil moisture while also analyzing crop images to detect diseases. It focuses on identifying early-stage diseases, calculating their severity percentage, and providing appropriate recommendations based on crop condition. Additionally, the project includes the development of a mobile application to display real-time data and send alerts to farmers. Overall, the objective is to improve decision-making, reduce crop loss, and enhance agricultural productivity through the use of advanced technology.

1.3 OBJECTIVE OF PROJECT

The main objective of this project is to design and develop a smart farming system that integrates IoT sensors, image processing, and AI techniques. The system aims to monitor environmental conditions such as temperature, humidity, and soil moisture while also analyzing crop images to detect diseases. It focuses on identifying early-stage diseases, calculating their severity percentage, and providing appropriate recommendations based on crop condition. Additionally, the project includes the development of a mobile application to display real-time data and send alerts to farmers. Overall, the objective is to improve decision-making, reduce crop loss, and enhance agricultural productivity through the use of advanced technology.

CHAPTER 3

BLOCK DIAGRAM



3.1 Block Diagram of IoT and AI Powered Smart Agriculture Monitoring System

CHAPTER 4

LITERATURE SURVEY

[1] The paper titled “Smart Agriculture System Using IoT with Help of AI-Techniques” by Krishna G. et al. (2026) presents a smart farming system that integrates Internet of Things (IoT) and Artificial Intelligence (AI) technologies to improve agricultural productivity. The system uses IoT sensors to monitor soil moisture, temperature, and humidity in real time, and the collected data is transmitted to a cloud server for storage and analysis. AI algorithms are applied to analyze environmental conditions and recommend suitable crops based on soil and seasonal factors. The system also uses image processing techniques with Convolutional Neural Networks (CNN) to detect plant diseases from crop images and suggest appropriate fertilizers and pesticides. In addition, a mobile application is developed to provide farmers with real-time data, crop recommendations, disease alerts, and information about government schemes. The proposed system helps in reducing manual effort, improving decision-making, and supporting sustainable farming. However, it has certain limitations such as dependency on internet connectivity, high initial cost, and the requirement of technical knowledge for operation.

[2] The paper “Modern tools for sustainable agriculture: a review of intelligent crop protection technologies” by Bilal Ahmad et al. reviews advanced technologies used in modern agriculture. The authors highlight the limitations of traditional crop protection methods in handling increasing food demand and climate challenges. The study focuses on intelligent crop protection using technologies such as IoT, AI, and Machine Learning. IoT devices like sensors and drones are used for real-time monitoring, while AI and ML help in early detection of pests and diseases. The paper also discusses the use of robotics, automation, remote sensing, and genome editing (CRISPR) to improve efficiency and crop resistance. The authors conclude that integrating these technologies helps in reducing chemical usage, saving resources, and improving productivity, leading to sustainable agriculture

[3] The paper titled “Development of an Integrated Smart Farm System for Open-Field Soybean Cultivation in Former Paddy Fields” by Chi-Yong An et al. presents a smart agriculture system for open-field farming. The system integrates IoT sensors for real-time monitoring, AI for pest and disease detection, UAV-based image analysis, and automated irrigation and drainage control. It also uses an Integrated Monitoring and Control System (IMCS) to manage all operations. The system was tested in a real field, showing an increase in soybean yield by about 20% and a reduction in labor by 53% compared to traditional methods. However, the system has limitations such as high cost, complexity, and the need for technical expertise.

[4] The paper titled “AI-Driven Smart Farming for Automated Plant Health Monitoring and Nutrient Deficiency Detection” by J. Serin et al. presents an intelligent farming system that combines IoT and Artificial Intelligence for continuous crop monitoring. The system uses sensors to measure soil moisture, temperature, and humidity, along with camera-based leaf image analysis using the DenseNet121 model to detect plant health and nutrient deficiencies. A microcontroller enables automated irrigation and cooling based on real-time data, and a mobile application provides alerts and monitoring. The model achieved an accuracy of 89%, outperforming traditional models. The system reduces manual effort and enables early detection of plant stress, but it may require proper setup, cost investment, and technical knowledge for effective use.

[5] The paper titled “A Review on Machine Learning-Based Precision Agriculture Techniques for Crop Farming Monitoring with IoT” by S. P. Sudha et al. presents a comprehensive review of smart farming techniques that integrate Machine Learning (ML) and the Internet of Things (IoT). The study explains how IoT sensors are used to collect real-time data such as soil moisture, temperature, and humidity, while ML algorithms are applied for crop health monitoring, disease detection, irrigation management, and yield prediction. It also provides a comparative analysis of different ML models based on accuracy and efficiency. The paper highlights key challenges such as data heterogeneity, sensor reliability, computational complexity, and cybersecurity issues. Furthermore, it discusses future improvements including hybrid ML models, explainable AI, and blockchain for secure and scalable smart agriculture system.

REFERENCES

- [1] Krishna G., Rajesh B., Dinesh B., Sravani B., Rajesh G., P. Dharnasi, and M. Sarvanan, "Smart Agriculture System Using IoT with Help of AI-Techniques," International Journal of Computer Technology and Electronics Communication (IJCTEC), vol. 9, no. 2, pp. 479487, Mar.-Apr. 2026 <https://ijctece.com/index.php/IJCTEC/article/view/476/429>
- [2] B. Ahmad et al., "Modern tools for sustainable agriculture: a review of intelligent crop protection technologies," Discover Agriculture, vol. 4, no. 19, 2026.
<https://doi.org/10.1007/s44279-025-00467-2>
- [3] C.-Y. An, C. Song, S. Song, S. Kim, H. Jeong, J. Ryu, and J. Park, "Development of an integrated smart farm system for open-field soybean cultivation in a former paddy fields," Front. Sustain. Food Syst., vol. 10, Art. no. 1702551, Jan. 2026, doi: 10.3389/fsufs.2026.
<https://doi.org/10.3389/fsufs.2026.1702551>
- [4] J. Serin et al., "AI-Driven Smart Farming for Automated Plant Health Monitoring and Nutrient Deficiency Detection," ASEAN Journal of Scientific and Technological Reports, vol. 29, no. 1, Dec. 2025, doi: 10.55164/ajstr.v29i1.258175.
<https://doi.org/10.55164/ajstr.v29i1.258175>
- [5] S. P. Sudha and J. B. Shajilin Loreto, "A review on machine learning-based precision agriculture techniques for crop farming monitoring with IOT," Discover Environment, vol. 4, no. 10, Jan. 2026, doi: 10.1007/s44274-025-8
<https://doi.org/10.1007/s44274-025-00305-8>